Band-1 Hadamard Order 2644 — Miyamoto (Paley-I H_660) + Paley-II Cross-Check; Gap 2052 Still OPEN

H_660/C_662/H_2644 PASS (Miyamoto + Paley-II); 2644 in SPK via Paley II q=1321; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
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Lab. `/workspace/experiments/band1/hadamard_2644_miyamoto/`.
Cutoff. 2026-09-15 (~01:37Z).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 660 and 2644 only** (verified $\max|HH^\top - nI| = 0$), via **two independent paths** for order 2644 (Miyamoto primary; Paley type II cross-check). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 2644 **enters SPK** via Paley II ($q = 1321$); it is **not** a Miyamoto-only bonus. Honest SPK audit: Paley-I **FAIL** ($q = 2643$ composite); Kronecker **FAIL** ($2644 = 4 \cdot 661$, 661 not Hadamard-admissible); Paley-II **PASS**. `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 2468 on box. This increment constructs the **preferred** next Miyamoto-eligible target after $q = 617$: **order 2644** ($2644 = 4 \cdot 661$, 661 prime $\equiv 1 \pmod 4$). Seed H_{660} via Paley type I ($q = 659$ prime $\equiv 3 \pmod 4$). Symmetric Paley conference C_{662} over prime 661. $H_{2644} = \text{Miyamoto}(H_{660}, C_{662})$. Independently, H_{2644} also verifies via **Paley type II** with $q = 1321$ (explains SPK membership). Unit tests **23/23 PASS**. SPK next gap remains **2052** (OPEN). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 2644; Miyamoto; Paley type I; Paley type II; Paley conference; constructive gap 2052; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in $\{\pm 1\}$ satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{660} constructed and verified (Paley type I, $q = 659$). 2. C_{662} constructed and verified (Paley conference, $q = 661$): $CC^\top = 661I$, zero diagonal. 3. H_{2644} constructed and verified (Miyamoto). **Also** constructed and verified via Paley type II ($q = 1321$). **Not** the Hadamard conjecture. 4. SPK: $2644 \in$ SPK (via Paley II). Paley-I and Kronecker paths for 2644 **rejected**. Next constructive gap remains **2052** (OPEN).

2. Arithmetic and eligibility

Quantity	Value
Target order	$2644 = 4 \cdot 661$
$q = 661$	prime, $661 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{660}
Needed conference	order $662 = q + 1$
H_{660} path	Paley type I, $q = 659$ (prime $\equiv 3 \pmod 4$)
Conference path	Symmetric Paley over prime 661
SPK path for 2644	Paley type II, $q = 1321$ prime $\equiv 1 \pmod 4$; order $2(q + 1) = 2644$
Paley-I for 2644	**REJECTED**: $q' = 2643$ not prime
Kronecker for 2644	**REJECTED**: only 4×661; $661 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{660}	Paley type I, $q = 659$	0	0.3648	**PASS**
C_{662}	Paley conference, $q = 661$	0	0.3391	**PASS**
H_{2644}	Miyamoto(H_{660}, C_{662})	0	8.1948	**PASS**
H_{2644} P-II	Paley type II, $q = 1321$	0	9.1095	**PASS**

Primary Miyamoto pipeline wall (incl. unit tests + SPK refresh + Paley-II cross-check) ≈ 18.9747 s. Compressed ± 1 matrices saved as `H660_pm1.npz`, `C662_pm1.npz`, `H2644_pm1.npz`, `H2644_paleyII_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible $\leq 2000 \cup$ on-box prior ships $\{24, 284, 508, 528, 540, 556, 568, 576, 600, 308, 616, 2004, 2036, 2116$ $\cup$ verified $H_{660} \cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4000). **Miyamoto is not folded into SPK**; order 2644 enters SPK **because** Paley II applies ($q = 1321$).

Quantity	Value
Known count	880
2644 in SPK	**True** (Paley II $q = 1321$)
660 in SPK	True (Paley I $q = 659$)
2468 in SPK	True (prior; Paley I $q = 2467$)
2404 in SPK	True (prior; Paley II $q = 1201$)
2276 in SPK	False (prior Miyamoto bonus)
Next standard gap	**2052** (OPEN)
Missing first (sample)	2052, 2060, 2076, 2092, 2108, 2148, 2156, 2172, 2212, 2284…

5. Gap 2052 (still OPEN; non-finding ONLY)

This ship does **not** fill gap 2052. Prior triage stands: Paley-I/II / Kronecker / Miyamoto-Paley-conf all **FAIL** for order 2052. Claim level for 2052 remains method triage / non-finding ONLY. Non-finding $\neq$ counterexample; Hadamard conjecture remains OPEN. No H_{2052} matrix is claimed.

6. Unit tests (23/23)

Test	Result
`H20_paley_I_control`	PASS
`H4_sylvester`	PASS

Test	Result
`paley_II_q5_H12` / `paley_II_q13_H28`	PASS
`miyamoto_H20_from_H4_conf6`	PASS
`paley_I_q11_H12`	PASS
`q661` / `n2644` / `q659` / `H660` / `C662` formulae	PASS
`q1321` / Paley-II order-2644 formula	PASS
`paley_I_H2644_path_rejected`	PASS
`kronecker_H2644_path_rejected`	PASS
`reject_all_ones_H8`	PASS
`conference_6_paley_control`	PASS
`miyamoto_eligible_2644_arith`	PASS
`conference_662_shape_expected`	PASS
`gap_2052_not_filled_by_this_ship`	PASS

7. Next optional targets

Miyamoto-eligible primes $q > 661$, $q \equiv 1 \pmod 4$ (sample):

q	$n = 4q$	Need H_{q-1}	On-box path notes
673	2692	672	Kronecker(4x168)
677	2708	676	PaleyII(q=337)
701	2804	700	PaleyII(q=349)
709	2836	708	PaleyII(q=353)
733	2932	732	no quick SPK seed noted
757	3028	756	no quick SPK seed noted

Also noted earlier: $q = 641 \to n = 2564$ (Kronecker 4×160) remains optional if desired. Fallback not needed this ship (2644 seed path succeeded).

Preferred next. Method-attack remaining SPK gap **2052** (GS/SDS for order ~513 if literature seeds appear), **or** construct H_{2692} / H_{2708} / H_{2564}, **or** Track B / Object 1 await. GATE-04 remains LOCKED.

8. Non-claims (explicit)

- Full Hadamard conjecture: **OPEN**
- Clay-closed / finished-ToE: **false / not claimed**
- Gap 2052 filled: **NOT claimed** (OPEN)
- GATE-04: **LOCKED**
- RH / P≠NP / YM / Frankl / NS-solved / AC-solved: **not claimed**

- Miyamoto-only bonus for 2644: **false** (in SPK via Paley II)

9. Reproducibility

```
cd /workspace/experiments/band1/hadamard_2644_miyamoto
python3 verify_hadamard_2644.py
# expects: Unit tests 23/23; H660/C662/H2644 Miyamoto PASS; Paley-II cross-check PASS
# artifacts: results.json, literature_ledger.json, SUMMARY.txt, *.npz
```

Lab path. `/workspace/experiments/band1/hadamard_2644_miyamoto/`
PDF shelf. `/workspace/DOWNLOAD_PDFs/2026-09-15_Band1_Hadamard_2644_Miyamoto.pdf`